

Consider a simple nonlinear system whose dynamics is described by the following first-order differential equation of the form

$$\dot{x} = f(t, x) \quad x(0) = x_0 \quad (1.1)$$

where $x \in \mathbb{R}^n$ is the state vector and x_0 is the state at time 0. And

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, \quad f = \begin{bmatrix} f_1(t, x_1, \dots, x_n) \\ f_2(t, x_1, \dots, x_n) \\ \vdots \\ f_n(t, x_1, \dots, x_n) \end{bmatrix} \in \mathbb{R}^n$$

One could expect that corresponding to each initial state x_0 .

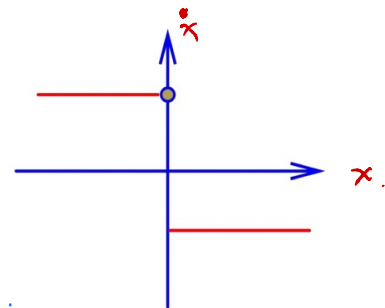
★ has at least one solution $\rightarrow$ existence of solutions.

★ has exactly one solution $\rightarrow$ uniqueness of solutions

★ has exactly one solution for all time $\rightarrow$ existence of a solution up to $t \rightarrow \infty$.

1. Lack of existence of solution

$$\dot{x} = -\text{sign}(x) \triangleq \begin{cases} -1 & x \geq 0 \\ 1 & x < 0 \end{cases}, \quad x(0) = 0.$$



Assume that there is a solution $x(t)$ which is continuous and starting from $x(0) = 0$.

Case I: $\exists t_1 > 0$, s.t. $x(t) < 0$, $\forall t \in (0, t_1)$

$$\dot{x} = 1, \quad x(t) - x(0) = t \Rightarrow x(t) = t > 0. \quad \text{Contradiction.}$$

Case II: $\exists t_1 > 0$, s.t. $x(t) \geq 0$, $\forall t \in (0, t_1)$.

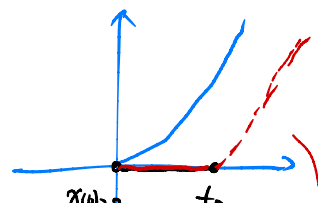
$$\dot{x} = -1, \quad x(t) - x(0) = -t \Rightarrow x(t) = -t < 0. \quad \text{Contradiction.}$$

2. Lack of uniqueness of solutions

$$\dot{x} = x^{1/3}, \quad x(0) = 0$$

i) $x(t) \equiv 0$

ii) $x(t) = \left(\frac{2}{3}t\right)^{\frac{3}{2}}$



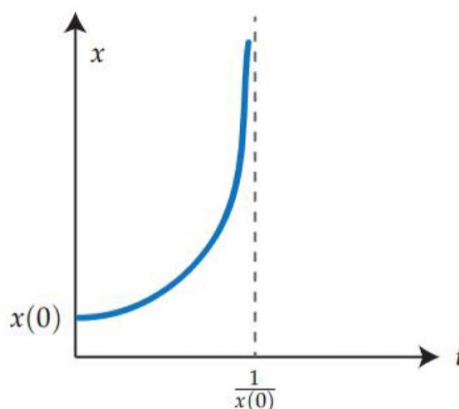
$x^{-\frac{1}{3}} \cdot \dot{x} = 1$
 $\frac{d\left(\frac{3}{2}x^{\frac{2}{3}}\right)}{dt} = 1$
 $\Rightarrow \frac{3}{2}x^{\frac{2}{3}} = t \Rightarrow x = \left(\frac{2}{3}t\right)^{\frac{3}{2}}$

$\begin{cases} \left(\frac{2}{3}(t-t_0)\right)^{\frac{3}{2}} & t > t_0 \\ 0 & t \leq t_0 \end{cases}$

3. Finite escape time

$$\dot{x} = x^2, \quad x(0) = 1$$

$$\begin{aligned} \frac{d}{dt}x^{-1} &= -x^{-2}\dot{x} = -1 \\ \Rightarrow \frac{1}{x(t)} - \frac{1}{x(0)} &= -t \\ \Rightarrow x(t) &= \frac{1}{\frac{1}{x(0)} - t} = \frac{1}{1-t} \end{aligned}$$

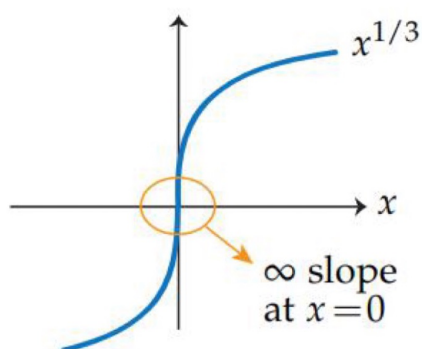


What are the requirements on the existence, uniqueness, and extension of the solution of (1.1)?

- If $f(\cdot)$ is continuous (C^0) then a solution exists, but C^0 is not sufficient for uniqueness.

Example: $\dot{x} = x^{\frac{1}{3}}$ with $x(0) = 0$

$$x(t) \equiv 0, \quad x(t) = \left(\frac{2}{3}t\right)^{\frac{3}{2}} \text{ are both solutions}$$



- Sufficient condition for uniqueness: "Lipschitz continuity" (more restrictive than C^0)

$$|f(x) - f(y)| \leq L|x - y|$$

$$|f(t, x) - f(t, y)| \leq L|x - y| \quad (*)$$

Definition: $f(\cdot)$ is *locally Lipschitz* if every point x^0 has a neighborhood where (*) holds for all x, y in this neighborhood and for all t for some L .

Example: $(\cdot)^{\frac{1}{3}}$ is NOT locally Lipschitz (due to ∞ slope)

$(\cdot)^3$ is locally Lipschitz: *not globally Lipschitz.*

$$x^3 - y^3 = \underbrace{(x^2 + xy + y^2)}_{\text{in any nbhd of } x^0, \text{ we can find } L \text{ to upper bound this}} (x - y)$$

in any nbhd
of x^0 , we can
find L to upper
bound this

$$\implies |x^3 - y^3| \leq L|x - y|$$

Preview of existence theorems:

1. $f(\cdot)$ is $C^0 \implies$ existence of solution $x(t)$ on finite interval $[0, t_f)$.
2. $f(\cdot)$ locally Lipschitz $\implies$ existence and uniqueness on $[0, t_f)$.
3. $f(\cdot)$ globally Lipschitz $\implies$ existence and uniqueness on $[0, \infty)$.

Let $f(t, x)$ be piecewise continuous in t and locally Lipschitz in x for all $t \geq t_0$ and all x in a domain $D \in \mathbb{R}^n$. Let W be a compact subset of D . Suppose that it is known that every solution of the system lies entirely in W . Then, there is a unique solution that is defined for all $t \geq t_0$.

Example: Pendulum

$$\ell m \ddot{\theta} = -k\ell\dot{\theta} - mg \sin \theta$$

Define $x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$. State space: $S^1 \times \mathbb{R}$.

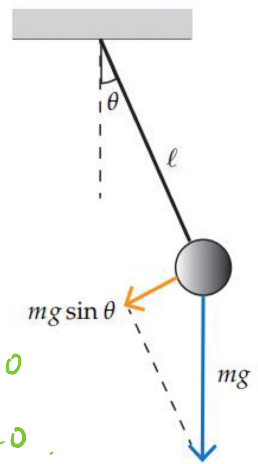
$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m}x_2 - \frac{g}{\ell} \sin x_1 \iff f_2(x_1, x_2) \equiv 0$$

$$\iff f_1(x_1, x_2) \equiv 0$$

$$x_2 = 0, \sin x_1 = 0 \Rightarrow x_1 = k\pi, k \in \mathbb{Z}$$



2. Mass-Spring System.

$$m\ddot{y} = F - F_{sp} - F_f$$

For a hardening spring, $F_{sp} = k(1 + a^2 y^2)y$
 $F_f = c \cdot \dot{y}$

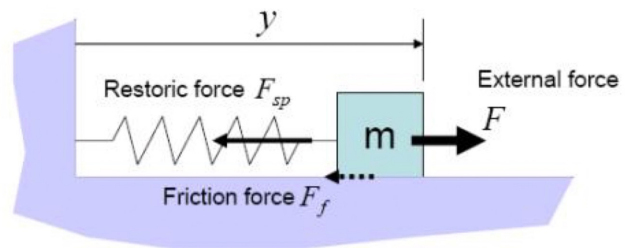
Assume $F = 0$, $m\ddot{y} = -k(1 + a^2 y^2)y - c\dot{y}$

Let $x_1 = y$, $x_2 = \dot{y}$. We have

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{c}{m}x_2 - \frac{k}{m}(1 + a^2 x_1^2)x_1$$



3. Negative-Resistance Oscillator.

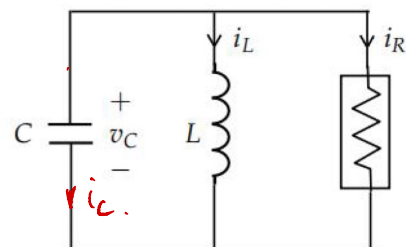
$$C \cdot \frac{dv_C}{dt} = -i_L - i_R$$

$$= -\frac{1}{L} \int_{-\infty}^t v_C(\tau) d\tau + v_C - v_C^3$$

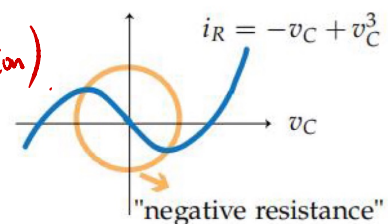
Differentiating once with respect to t ,

$$LC \cdot \ddot{v}_C + L(3v_C^2 - 1)\dot{v}_C + v_C = 0$$

Let $x_1 = v_C$, $x_2 = \dot{v}_C$. We have

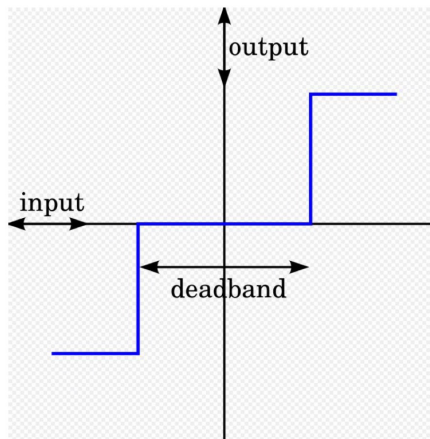
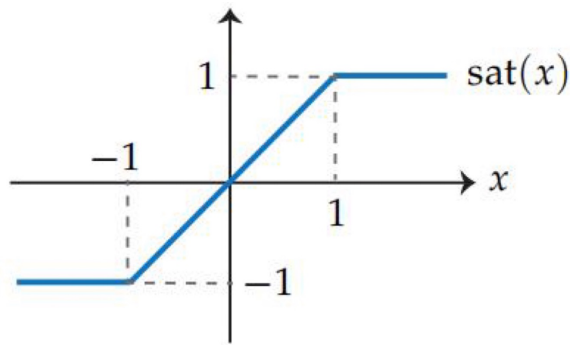
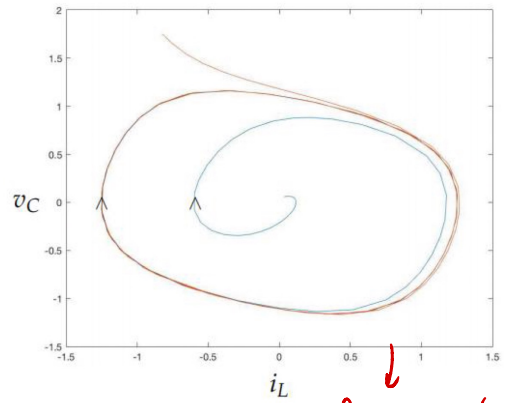


(Van der Pol Equation)



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{L}{C} (3x_1^2 - 1) x_2 - \frac{1}{LC} x_1$$



(平衡点)

Definition: Equilibrium Points

$$\dot{x} = f(t, x)$$

x^* is said to be an equilibrium point of (1.1) if and only if $f(t, x^*) \equiv 0, \forall t \geq t_0$.

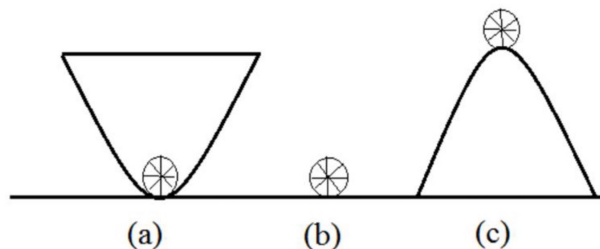
In other words, $x = x^*$ is an equilibrium point, whenever the state of the system starts at x^* , it will remain at x^* for all future time.

In the linear case: $\dot{x} = Ax$ ($Ax = 0$)

- A unique solution $x = 0$, when A is nonsingular.
- A continuum of solutions, when A is singular.

Definition: Isolated Equilibrium

An equilibrium point x^* is isolated if there exists some $\delta > 0$, such that there is no other equilibrium points in the ball $B(x_0, \delta) \triangleq \{x \mid \|x - x^*\| \leq \delta\}$.



$$\dot{x} = f(x) \quad f: \mathbb{R}^n \rightarrow \mathbb{R}^n \quad \text{time-invariant (autonomous)} \quad \rightarrow \text{自治系统.}$$

$$\dot{x} = f(t, x) \quad f: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n \quad \text{time-varying (non-autonomous)} \quad \downarrow \text{非自治系统.}$$

— For autonomous systems,

$$\dot{x} = f(x), \quad x(0) = x_0.$$

Let $y(t) = x(t-t_0)$. We have $y(t_0) = x_0$. / have the same solution.

Then $y = \frac{dx(t-t_0)}{dt} = f(x(t-t_0)) = f(y), \quad y(t_0) = x_0$ ↓

This property is called shift-invariant.

— For non-autonomous systems

$$\dot{x} = f(t, x), \quad x(0) = x_0.$$

Let $y(t) = x(t-t_0)$. We still have $y(t_0) = x_0$. / have different solutions.

However, $y = \frac{dx(t-t_0)}{dt} = f(t-t_0, x(t-t_0)) = f(t-t_0, y) \neq f(t, y)$

The initial time matters, and the shift-invariant property does not hold.

That's why we need to include t_0 in the definitions of stability analysis for non-autonomous systems and introduce the concept of uniformity.